

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Summer Examination – 2024 Course: B. Tech. Branch: Computer and Allied Branches Semester: III Subject Code & Name: BTCOC305B Object Oriented Programming in Java Max Marks: 60 Date: 09/07/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/ CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	What are the features of Java Programming Language? Explain in brief	2/CO1	6
B)	Define Class, Method, and Object. Give the syntax to define these in Java.	2/CO1	6
C)	Discuss various types of Constructors in Java.	2/CO1	6
Q.2	Solve Any Two of the following.		12
A)	Explain a) Conditional Operators b) Bitwise Operators c) Logical Operators in Java	2/CO2	6
B)	Discuss the scope of variables with suitable examples. Explain with the example "For..Each" loop statement provided in Java.	2/CO2	6
C)	What is a method? Explain with Java syntax how a method is defined. What do you mean by static methods in Java?	2/CO2	6
Q. 3	Solve Any Two of the following.		12
A)	How can we create a single-dimensional array initialized with random values in Java? What do you mean by ragged array? How is it created using Java? Explain with an example.	2/CO3	6
B)	i) Write a Java code for binary search searching technique. ii) Discuss the "arraycopy" method of the System class.	3/CO3	6
C)	i) Demonstrate with example how we use an array to pass the variable number of parameters to a method. ii) Explain with suitable example how single dimensional arrays are passed in methods?	3/CO3	6
Q.4	Solve Any Two of the following.		12
A)	What is an abstract class in java? What is an interface? List the rules to create an interface in java with example.	2/CO4	6
B)	Discuss how various types of inheritance in Java.	2/CO4	6
C)	Write a short note on Polymorphism for Java.	2/CO4	6
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Q. 5	Solve Any Two of the following.		12
A)	What is an exception? What are the types of exceptions in Java? Describe the try-catch construct	2/CO5	6
B)	Discuss throw, throws and finally statements in Java.	2/CO5	6
C)	What is package? Write a program to create user defined package in java.	3/CO5	6
*** End ***			